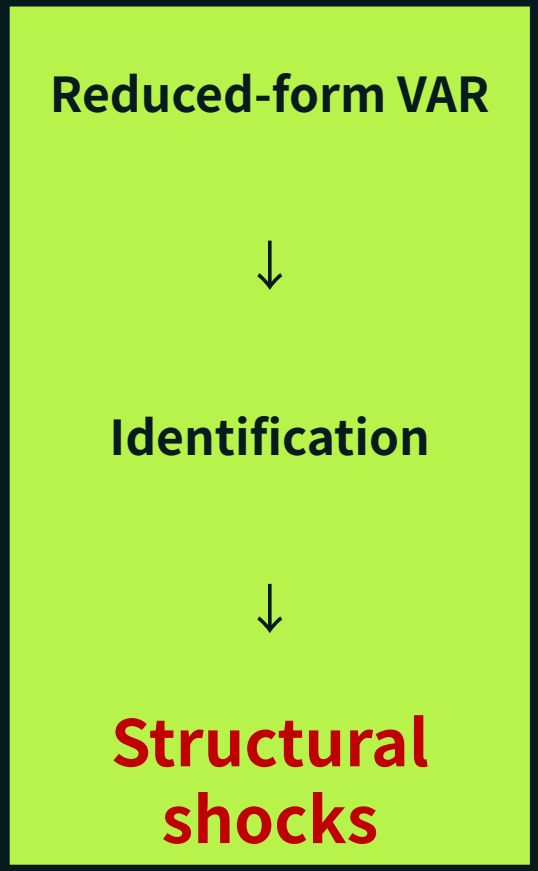
VAR and Structural VAR Identification Matters

Why the same data can yield different economic conclusions



Financial Time Series · 06

Four steps take us from the reduced form to the structural form

01		**Why SVAR**	**One correlation, three stories** Reduced form · forecast errors · shocks
02		**The identification problem**	**Unmixing the shocks** Counting restrictions · one S, two Ps · ordering
03		**Identification methods**	**Six families of assumptions** Recursive · long-run · sign · proxy · variance
04		**Finance cases and labs**	**Assumptions change conclusions** Monetary policy · oil · stocks · Claude Code

Only one thing differs: the identifying assumption

A reduced-form VAR explains each variable by the past of all variables

Basics · Reduced-form VAR — the form estimated directly from data

Reduced-form VAR

$$z_t = c + A_1 z_{t-1} + \cdots + A_p z_{t-p} + e_t, \qquad \mathrm{Cov}(e_t) = S$$

Reduced-form shock · forecast error

What the past cannot explain

Correlated across equations, unnamed

Structural shock

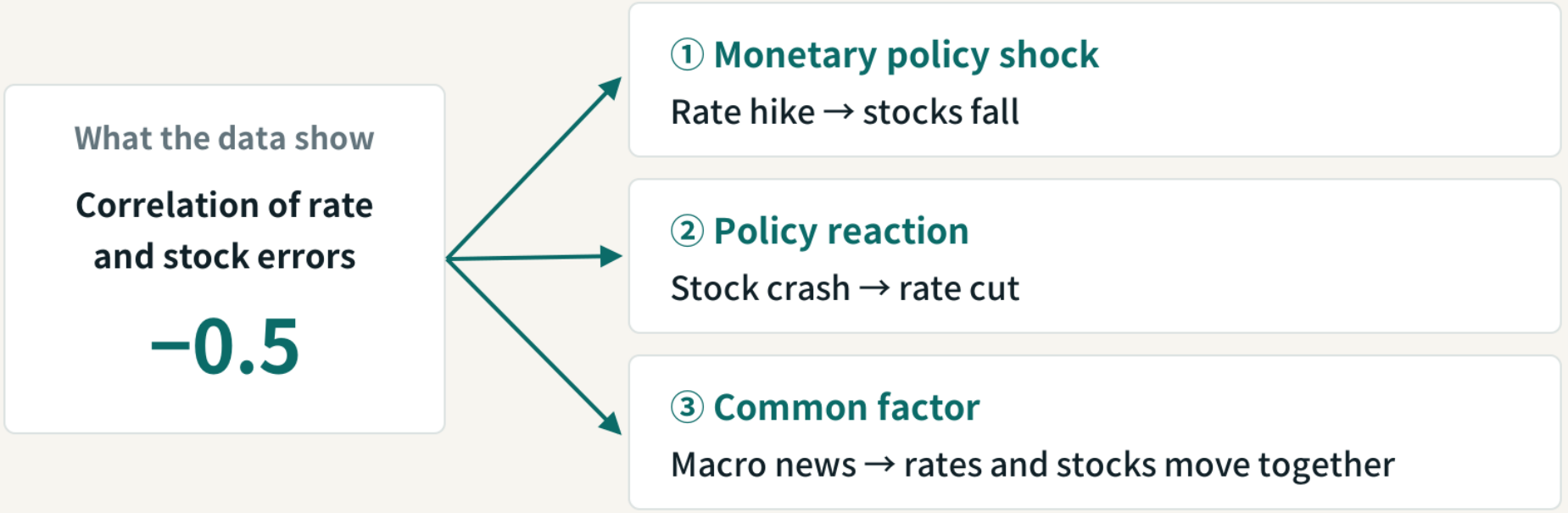
A named shock: policy, demand

Mutually uncorrelated, unit variance

VAR errors are just the unpredicted part; by themselves they carry no name

A single correlation of −0.5 fits all three stories

Errors of the rate equation and the stock equation move together on the same day

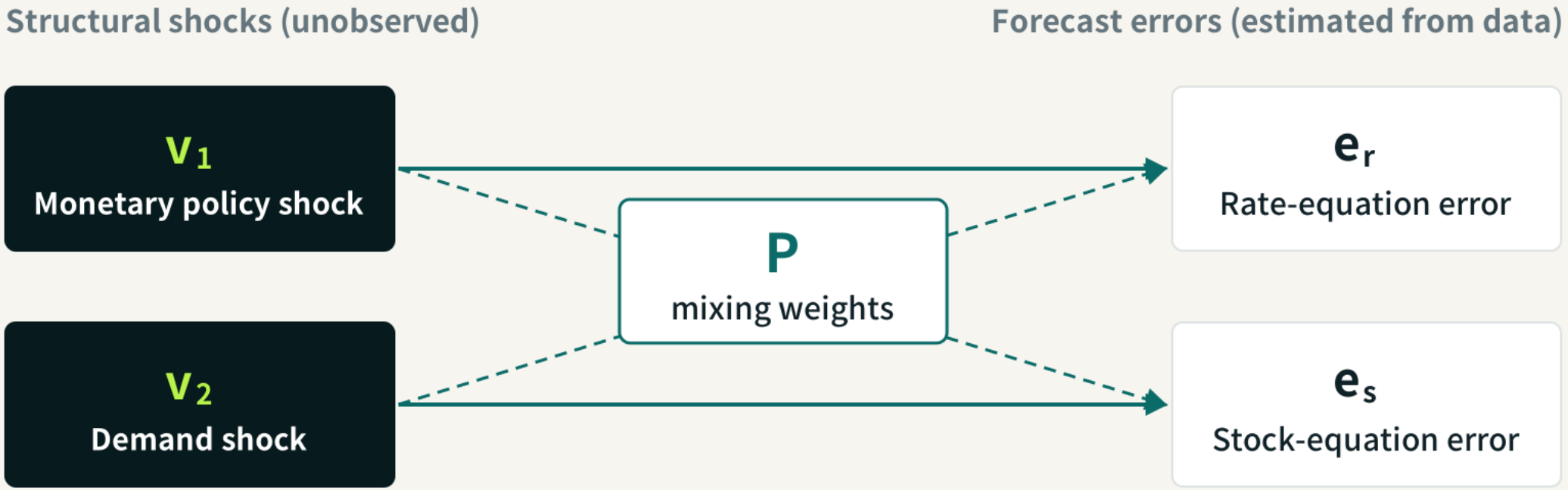


A reduced-form VAR cannot tell the three stories apart

An SVAR states the assumption that picks one

Forecast errors are mixtures of structural shocks

Solid = share going to its own equation, dashed = share leaking into the other



The data only show the right side (forecast errors)

Recovering the left side (structural shocks) needs an assumption that fixes P

A single matrix P links structural shocks to forecast errors

Structural shocks are normalized to be uncorrelated with unit variance

Link to structural shocks

$$e_t = P\,v_t, \qquad \mathrm{Cov}(v_t) = I \implies S = PP'$$

Each column of P

One shock's impact

Same-period responses

What the data give

Only the covariance S

P is not observed

Once P is fixed

Shocks get names

IRF, FEVD by shock

Every SVAR exercise ultimately comes down to choosing P

The structural form writes contemporaneous relations as a matrix

Reference · general form — premultiplying by the inverse gives the reduced form

Structural VAR (contemporaneous matrix on the left)

$$B_0\, z_t = c + B_1 z_{t-1} + \cdots + B_p z_{t-p} + v_t$$

Mapping to the reduced form

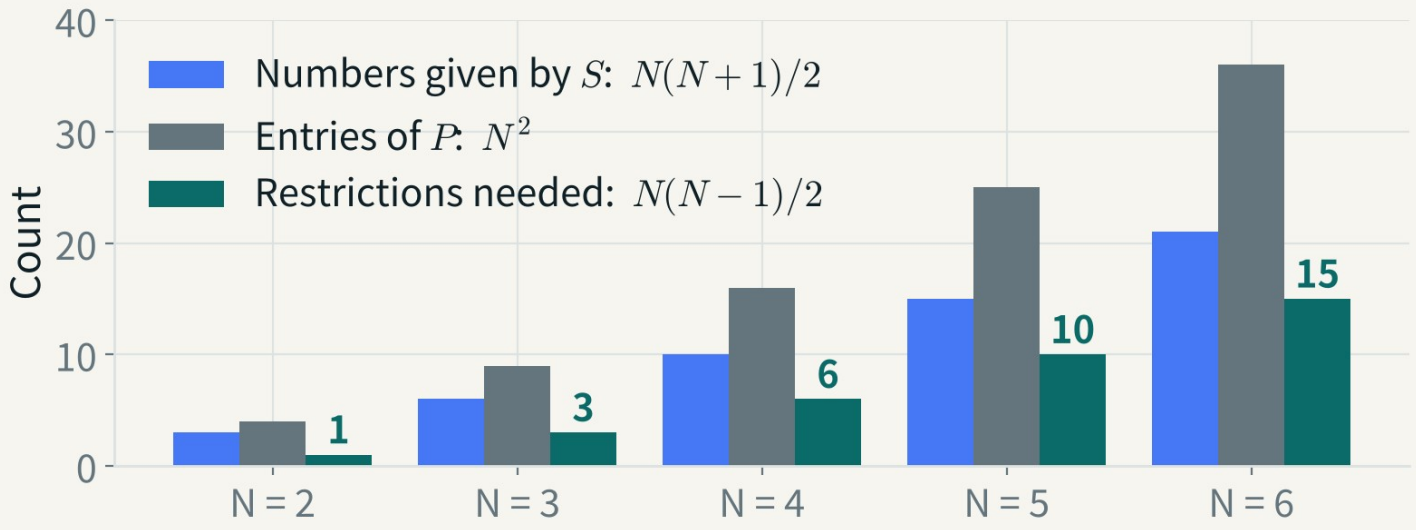
$$A_i = B_0^{-1} B_i, \qquad e_t = B_0^{-1} v_t \implies P = B_0^{-1}$$

※ The reduced form gives only the A matrices and S — B0 (hence P) is not pinned down by the data

Structural form or P alike, the hidden contemporaneous relations must come from assumptions

Identifying restrictions grow quickly as variables are added

S is symmetric, so it has few distinct numbers, while every entry of P must be filled



$$N^2 - \frac{N(N+1)}{2} = \frac{N(N-1)}{2}$$

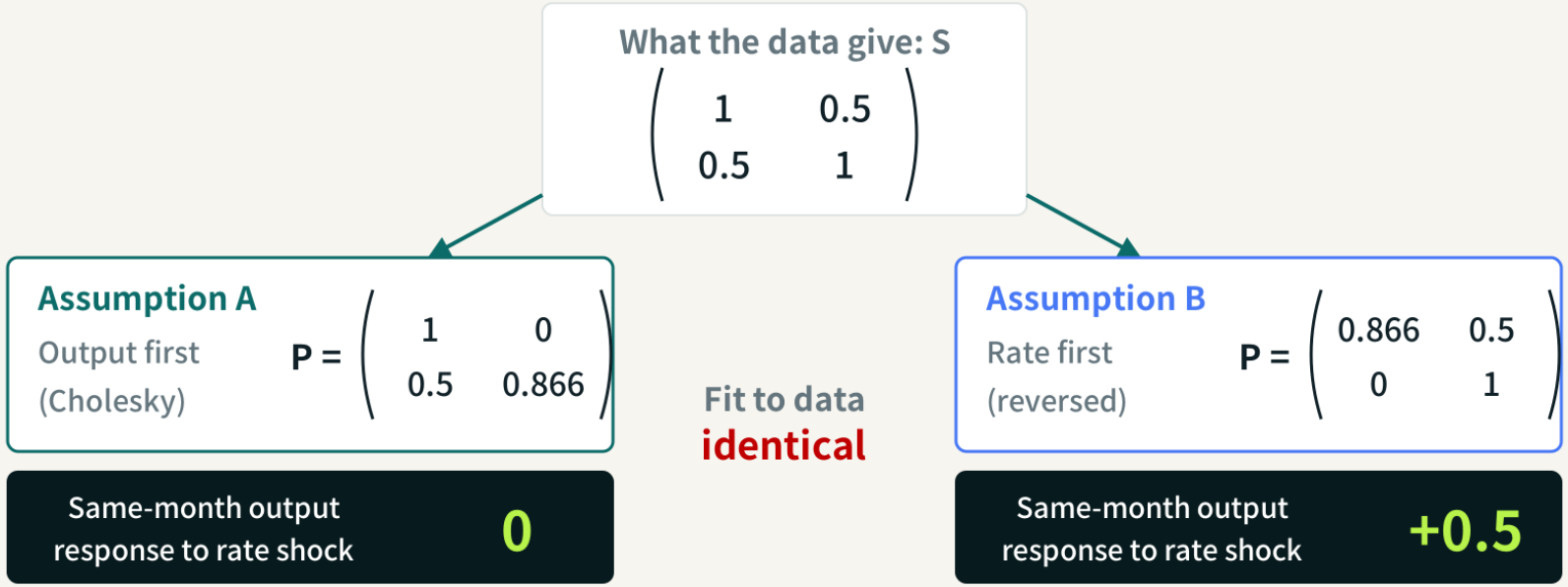
Restrictions needed

2 variables → 1
3 variables → 3
4 variables → 6

The missing information must come from economic reasoning, not from the data

Two assumptions give different impact responses from the same data

Two variables, output (y) and interest rate (x), with the same reduced-form covariance S



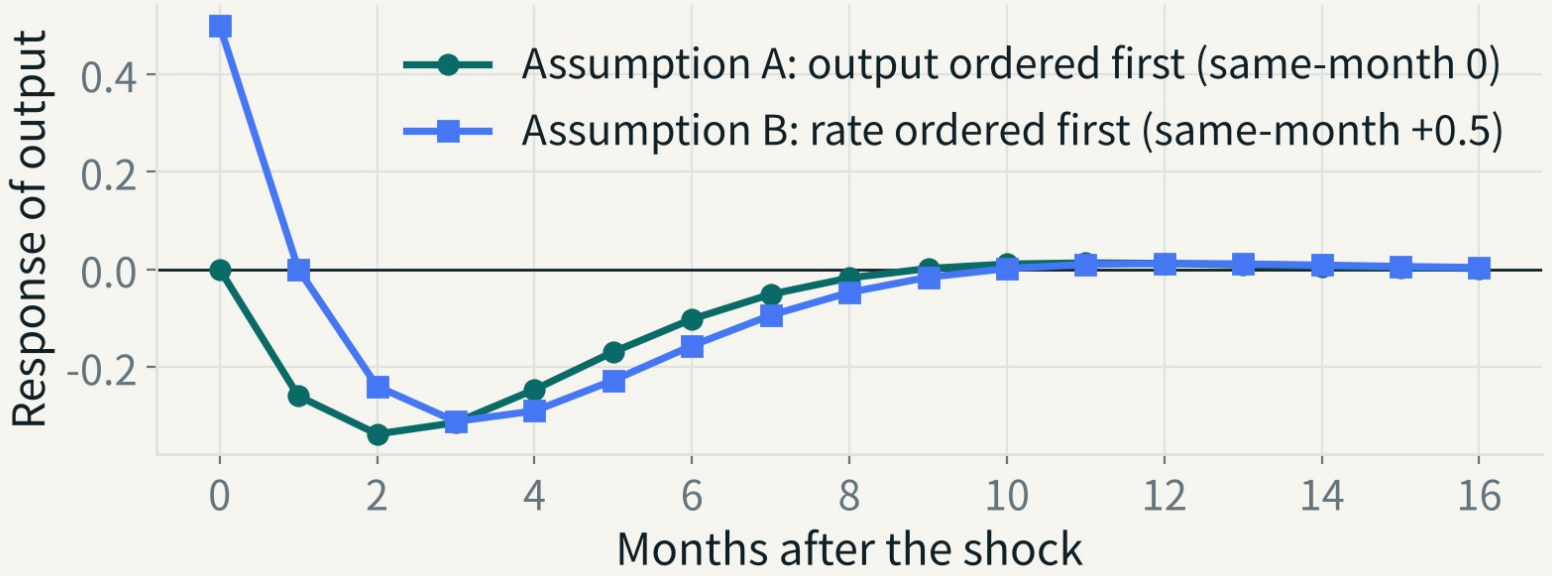
※ Both Ps reproduce S exactly — the hand check by matrix multiplication is in the workbook

The data cannot tell which one is right

The difference in conclusions comes entirely from the assumption

Changing only the variable ordering changes the impulse response path

Response of output to a one-standard-deviation rate shock under two recursive orderings



※ Simulation — S from the previous slide, lag coefficients illustrative

The Cholesky ordering is an economic assumption, not a computational option

Identification methods fall into six families by the type of assumption

Top three rows use zero restrictions; bottom three (teal bar) do not

Method	What it assumes	Key studies	Cost (weakness)
Recursive (Cholesky)	Later shocks: 0 impact on earlier variables	Sims · CEE	Order changes results
Short-run zeros	Zeros only where theory says	Sims–Zha	Needs theory per zero
Long-run	Zero in the long-run cumulative effect	Blanchard–Quah	Fragile LR estimates
Sign restrictions	Only the signs of responses	Uhlig	Answer is a range
External proxy	Shock measured outside the VAR	Gertler–Karadi	Good proxies are rare
Heteroskedasticity	Shock variance differs by regime	Rigobon–Sack	Regime split assumed

Weaker assumptions give blurrier answers; stronger ones need more justification

Every method has a cost

Long-run restrictions zero out one cumulative-effect entry

Reference · Blanchard–Quah (1989), two variables: output growth and unemployment

Long-run cumulative effect of shocks

$$L = \left(I - A_1 - \cdots - A_p\right)^{-1} P$$

0 = demand shock → output level

$$L = \begin{pmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{pmatrix}$$

The assumption

Demand: no long-run output effect

Short-run responses are left free

Weakness

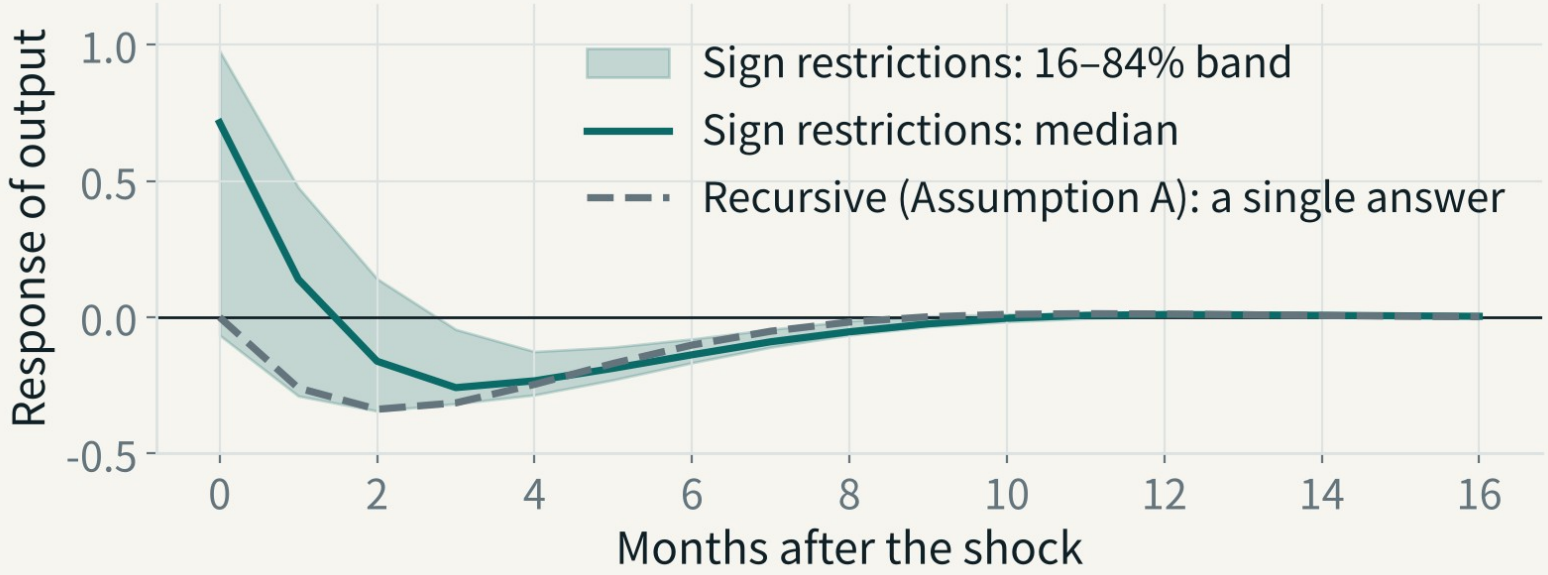
Sensitive to long-run estimates

Imprecise estimates swing results

Restricting the long run instead of the short run lets the data speak on short-run dynamics

Sign restrictions return a range of answers instead of a single one

Only the rate response is restricted to be positive for h = 0–3; the output response is left free



※ Simulation — 4,000 rotation draws; the band uses the 1,642 that satisfy the signs (seed 7)

Weaker than zero restrictions, hence more credible,

but even the sign of the impact output response may stay undetermined

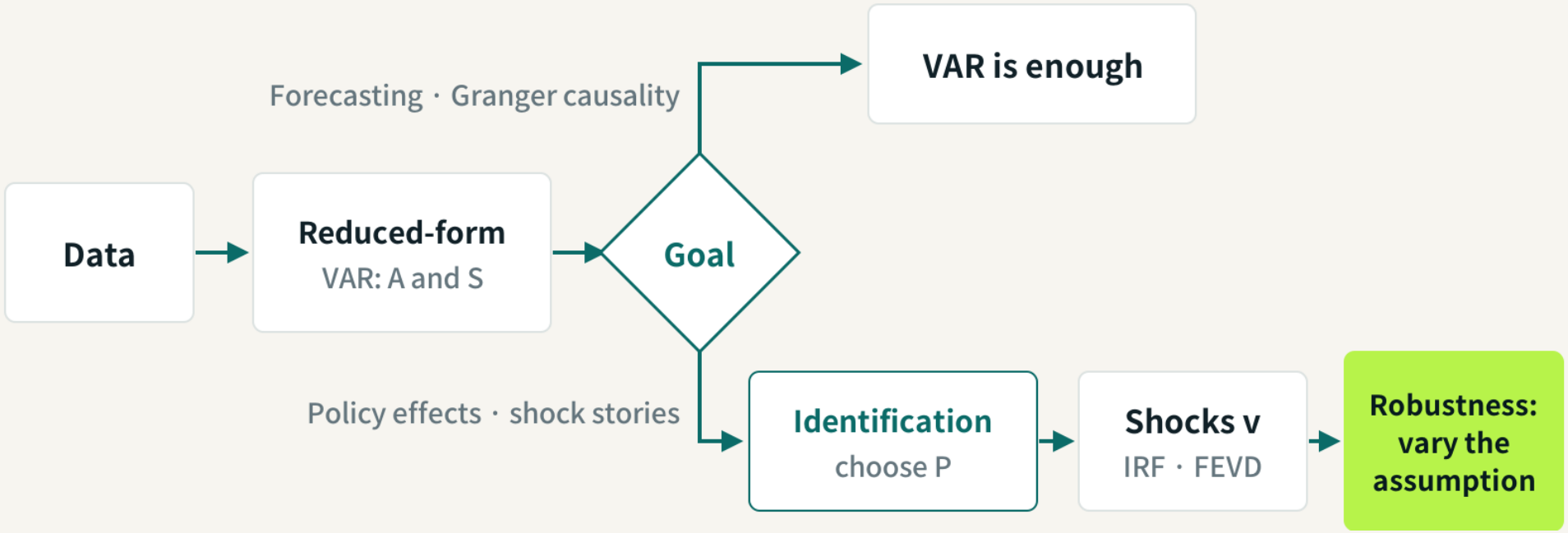
The reduced form forecasts; the structural form interprets

Item	Reduced-form VAR	Structural VAR
Purpose	Forecasting · Granger causality	Policy effects · shock propagation
Shocks	Forecast errors · correlated	Named shocks · uncorrelated
Assumptions	Lag length · stationarity	+ identifying assumptions
Uniqueness	**Uniquely determined**	**Changes with the assumption**
Reporting	Forecast accuracy · test tables	Identification + IRF + robustness

Always report SVAR results together with the identifying assumption used

The analysis splits into two paths depending on the goal

Reduced-form estimation is the common start; identification is added only for interpretation



Once you choose an identification, always vary it to check robustness

There are three common misconceptions about SVARs

Top line = misconception, bullets = correction

Misconception 1

Cholesky is neutral

- A recursive SVAR
- Order = assumption
- Justify the order

Misconception 2

SVAR is always better

- For forecasting, a VAR is enough
- Identification adds no forecast power

Misconception 3

Fits theory, so valid

- Sign restrictions assume results
- This can become circular

Plausible results and valid identification are two different things

How sharp policy effects look depends on the identification

Case · Christiano–Eichenbaum–Evans (1999) vs Uhlig (2005)

CEE · recursive

Output falls for quarters

- Order: output, prices → rate
- Hike → prices rise: price puzzle

A sign of incomplete identification

Uhlig · sign restrictions

No clear output response

- Tightening: prices, money not up
- Output response left free

Weaker assumptions, weaker conclusions

Even "rate hikes slow the economy" rests on an identifying assumption

Oil price shocks can be interpreted only after splitting them by source

Case · Kilian (2009), recursive ordering: oil production → global real activity → oil price



One oil price rise hits stocks differently under supply cuts and demand booms

A single reduced-form "oil shock" cannot support investment decisions

Heteroskedasticity identifies instantly reacting rates and stocks

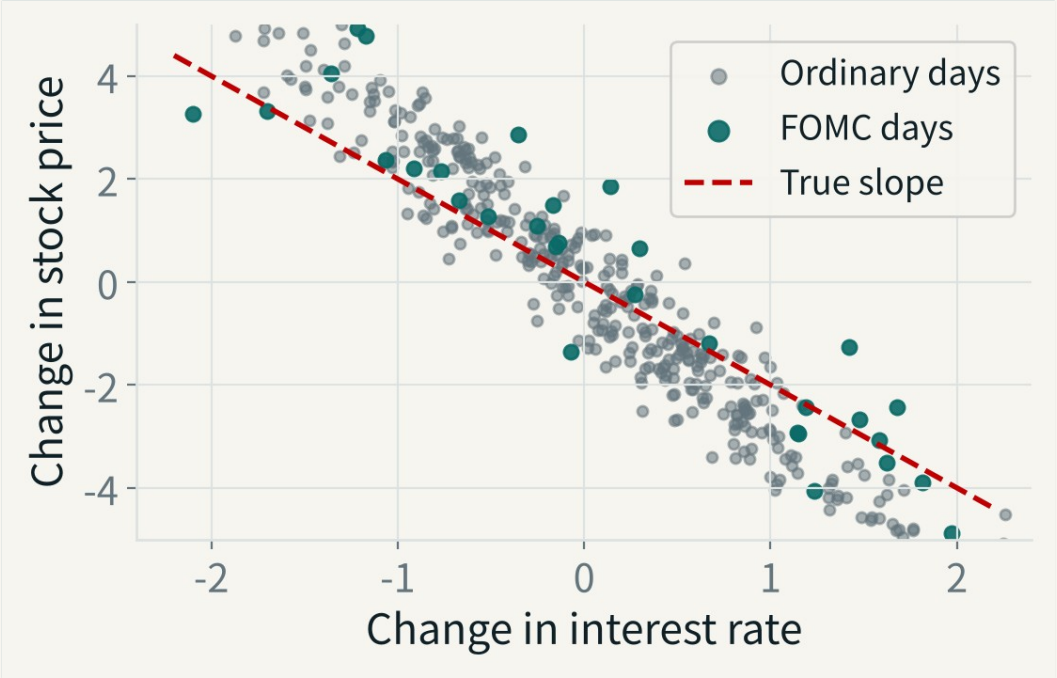
Case · Rigobon–Sack (2004), exploiting high rate volatility on FOMC days

How to read

- FOMC days: bigger shocks
- They trace the true slope

※ Simulated data (seed 11)

Rate hike → stocks fall



Recursive assumptions do not suit financial variables that react to each other instantly

External proxies and long-run restrictions need no ordering

Case · Gertler–Karadi (2015) proxy SVAR, Blanchard–Quah (1989) long-run restrictions

Gertler–Karadi · external proxy

Spreads carry the shock

- Proxy: futures moves at FOMC
- Policy shocks widen spreads

Market data replace the ordering

Blanchard–Quah · long run

Permanent vs transitory

- Output swings → two shocks
- Pension long-run growth inputs

Separates permanent from transitory

Asset allocation scenarios start by asking whether a shock is permanent or transitory

Lab 1 — Apply three identifications to the same data and compare

FRED monthly data 1965–2007 · IRFs to a one-standard-deviation rate hike

Download 4 FRED series → **Log transform BIC lags · S** → **Identify 3 ways IRF 48 months** → **Overlay plots sign band** → **5-line md save · open**

Three identifications

Output first · rate first · sign

What to check

How sensitive the conclusion is

Students paste the prompt and only check the results

Downloading, running and fixing errors are all left to Claude Code

Lab 2 — Separate supply and demand shocks with long-run restrictions

FRED quarterly data · Blanchard–Quah SVAR of real GDP growth and unemployment

GDP · unemp. quarterly data → Growth rates ADF · KPSS → Long-run SVAR identified → Supply, demand IRF 40 qtrs → Hist. decomp. 2008 · 2020

Built from scratch

Long-run cumulative effect matrix

What to check

Which shock drove each recession

If a library lacks the feature, let Claude Code implement it

Receive svar_blanchard_quah.md and plots, and just interpret them

Same data, same fit.

Different identification,

different economics.

When reading SVAR results,
check the identification first.

※ Practice check — state the identification in one sentence and show results under other orderings. For rates, stocks and FX, try sign, proxy or heteroskedasticity first.